

Obesity and addiction: neurobiological overlaps.



Drug addiction and obesity appear to share several properties. Both can be defined as disorders in which the saliency of a specific type of reward (food or drug) becomes exaggerated relative to, and at the expense of others rewards. Both drugs and food have powerful reinforcing effects, which are in part mediated by abrupt dopamine increases in the brain reward centres. The abrupt dopamine increases, in vulnerable individuals, can override the brain's homeostatic control mechanisms. These parallels have generated interest in understanding the shared vulnerabilities between addiction and obesity. Predictably, they also engendered a heated debate. Specifically, brain imaging studies are beginning to uncover common features between these two conditions and delineate some of the overlapping brain circuits whose dysfunctions may underlie the observed deficits. The combined results suggest that both obese and drug-addicted individuals suffer from impairments in dopaminergic pathways that regulate neuronal systems associated not only with reward sensitivity and incentive motivation, but also with conditioning, self-control, stress reactivity and interoceptive awareness. In parallel, studies are also delineating differences between them that centre on the key role that peripheral signals involved with homeostatic control exert on food intake. Here, we focus on the shared neurobiological substrates of obesity and addiction. © 2012 The Authors. *obesity reviews* © 2012 International Association for the Study of Obesity.